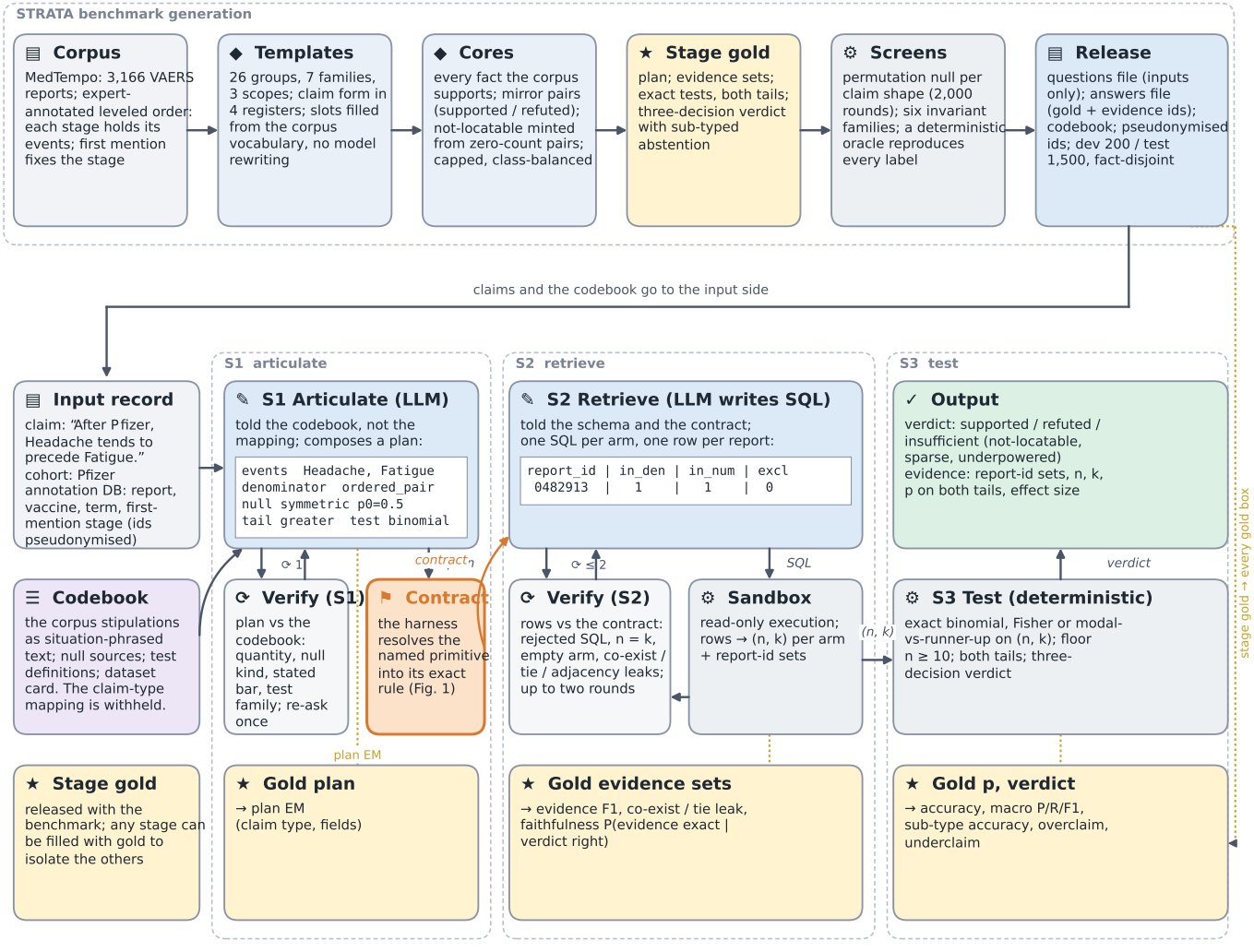

STRATA benchmark generation
Corpus
MedTempo: 3,166 VAERS reports; expert-annotated leveled order: each stage holds its events; first mention fixes the stage
Templates
26 groups, 7 families, 3 scopes; claim form in 4 registers; slots filled from the corpus vocabulary, no model rewriting
Cores
every fact the corpus supports; mirror pairs (supported / refuted); not-locatable minted from zero-count pairs; capped, class-balanced
Stage gold
plan; evidence sets; exact tests, both tails; three-decision verdict with sub-typed abstention
Screens
permutation null per claim shape (2,000 rounds); six invariant families; a deterministic oracle reproduces every label
Release
questions file (inputs only); answers file (gold + evidence ids); codebook; pseudonymised ids; dev 200 / test 1,500, fact-disjoint
claims and the codebook go to the input side
S1 articulate
S2 retrieve
S3 test
Input record
claim: "After Pfizer, Headache tends to precede Fatigue."
cohort: Pfizer
annotation DB: report, vaccine, term, first-mention stage (ids pseudonymised)
S1 Articulate (LLM)
told the codebook, not the mapping; composes a plan:
events Headache, Fatigue
denominator ordered_pair
null symmetric p0=0.5
tail greater test binomial
S2 Retrieve (LLM writes SQL)
told the schema and the contract; one SQL per arm, one row per report:
report_id | in_den | in_num | excl
0482913 | 1 | 1 | 0
Output
verdict: supported / refuted / insufficient (not-locatable, sparse, underpowered)
evidence: report-id sets, n, k, p on both tails, effect size
Codebook
the corpus stipulations as situation-phrased text; null sources; test definitions; dataset card. The claim-type mapping is withheld.
Verify (S1)
plan vs the codebook: quantity, null kind, stated bar, test family; re-ask once
Contract
the harness resolves the named primitive into its exact rule (Fig. 1)
Verify (S2)
rows vs the contract: rejected SQL, n = k, empty arm, co-exist / tie / adjacency leaks; up to two rounds
Sandbox
read-only execution; rows → (n, k) per arm + report-id sets
S3 Test (deterministic)
exact binomial, Fisher or modal-vs-runner-up on (n, k); floor n ≥ 10; both tails; three-decision verdict
contract
SQL
verdict
(n, k)
plan EM
Stage gold
released with the benchmark; any stage can be filled with gold to isolate the others
Gold plan
→ plan EM (claim type, fields)
Gold evidence sets
→ evidence F1, co-exist / tie leak, faithfulness P(evidence exact | verdict right)
Gold p, verdict
→ accuracy, macro P/R/F1, sub-type accuracy, overclaim, underclaim
stage gold → every gold box